

1.Git/Github Output:-

The screenshot shows a GitHub repository page for 'Smart_Education_Analytics' by user 'NareshDesale'. The repository is public and has 8 commits. The commit history is as follows:

Commit	Message	Time Ago
86f584c	Add GitHub Actions CI/CD pipeline	32 minutes ago
	Initial project setup	10 hours ago
	Add project folder structure	9 hours ago
	Add Kubernetes deployment and service YAML files	40 minutes ago
	Move notebook into notebooks folder	10 hours ago
	Add project folder structure	9 hours ago
	Add Dockerfile and test Spark application in Docker	7 hours ago
	Move notebook into notebooks folder	10 hours ago
	Add Dockerfile and test Spark application in Docker	7 hours ago
	Add project folder structure	10 hours ago
	Add Dockerfile and test Spark application in Docker	7 hours ago

The repository contains the following files and folders:

- .github/workflows
- data
- documentation
- kubernetes
- notebooks
- screenshots
- src
- .gitignore
- Dockerfile
- README.md
- requirements.txt

The right sidebar shows the repository's metadata:

- About:** No description, website, or topics provided.
- Readme:** Link to the README file.
- Activity:** Link to the repository's activity.
- Stars:** 0 stars.
- Watching:** 0 watching.
- Forks:** 0 forks.
- Releases:** No releases published. [Create a new release](#).
- Packages:** No packages published. [Publish your first package](#).
- Contributors:** 1 contributor: NareshDesale.

2.Docker Output:-

The image displays two screenshots of a Visual Studio Code editor window, showing a workflow for setting up a Docker environment. The workflow is defined in a `ci-cd.yml` file, and the terminal shows the execution of `docker` commands.

Workflow Steps (ci-cd.yml):

- name:** Show Project Structure
run: `ls -R`
- name:** Set up Python
uses: `actions/setup-python@v5`
with: `python-version: "3.10"`
- name:** Install Python Dependencies
run: `python -m pip install --upgrade pip`
`pip install -r requirements.txt`
- name:** Run Basic Python Test

Docker Images (Terminal Output):

PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> `docker --version`
Docker version 29.6.1, build 8900f1d

PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> `docker images`

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
docker/desktop-kubernetes-kubernetes-v1.36.1-cni-v1.9.1-critools-v1.35.0-cri-dockerd-v0.3.25-1-debian	1c21deb374b	594MB	188MB	
docker/desktop-storage-provisioner:v4.0	ca46f2a97ef4	82.1MB	24.7MB	U
docker/desktop-vmkit-controller:v4.0	8cb4e79ec634	52.6MB	11.9MB	U
kindest/node:v1.36.1	3489c7674813	1.31GB	383MB	
registry.k8s.io/coredns/coredns:v1.14.2	e7e6440cf1d1e	103MB	23.3MB	U
registry.k8s.io/etcd:3.6.8-0	397189418d1a	88.8MB	22.9MB	U
registry.k8s.io/kube-apiserver:v1.36.1	2460df74307c	128MB	29MB	U
registry.k8s.io/kube-controller-manager:v1.36.1	c8adb8dae5b	114MB	25.7MB	U

Docker Images (Terminal Output - Continued):

registry.k8s.io/kube-proxy:v1.36.1	a96b6e12863e	111MB	28.9MB	U
registry.k8s.io/kube-scheduler:v1.36.1	3a4333e8ba6e	82.7MB	19.7MB	U
smart-education-analytics:latest	f548e0e8e3dc	1.06MB	318kB	U
9cf47aa264b4	9cf47aa264b4	2.78GB	924MB	U

PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics>

```
File Edit Selection View Go Run Terminal Help Smart_Education_Analytics
EXPLORER kubernet... .gitkeep output .gitkeep screenshots .gitkeep src requirements.txt deployment.yaml service.yaml ci-cd.yml .gitignore
SMART EDUCATION ANAL...
  .github\workflows
  ! ci-cd.yml
  > data
  > documentation
  .gitkeep
  kubernet...
  .gitkeep
  ! deployment.yaml
  ! service.yaml
  ! notebooks
  Case_Study(AI).ipynb
  > output
  .gitkeep
  > screenshots
  .gitkeep
  > src
  .gitkeep
  > main.py
  .gitignore
  Dockerfile
  README.md
  requirements.txt

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> docker run --rm smart-education-analytics
Using Spark's default log4j profile: org/apache/spark/log4j2-defaults.properties
Setting default log level to "WARN".
To adjust logging level use sc.setLogLevel(newLevel). For SparkR, use setLogLevel(newLevel).
26/07/03 04:22:56 WARN NativeCodeLoader: unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Spark Started Successfully
Spark Version: 4.1.2
xAPI Dataset Records: 480

[gender|Nationality|PlaceofBirth|StageID|GradeID|SectionID|Topic|Semester|Relation|raisedhands|VisitedResources|AnnouncementsView|Discussion|ParentAnsweringSurvey|Parentscho
Satisfaction|StudentAbsenceDays|Class]

+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| M | Good | Kuwait | lowerlevel | G-04 | A | IT | F | Father | 15 | 16 | 2 | 20 | Yes |
| M | Good | Kuwait | lowerlevel | G-04 | A | IT | F | Father | 20 | 20 | 3 | 25 | Yes |
| M | Good | Kuwait | lowerlevel | G-04 | A | IT | F | Father | 10 | 7 | 0 | 30 | No |
| M | Bad | Kuwait | Above-7 | G-04 | A | IT | F | Father | 30 | 25 | 5 | 35 | No |
```

```
File Edit Selection View Go Run Terminal Help Smart_Education_Analytics
EXPLORER kubernet... .gitkeep output .gitkeep screenshots .gitkeep src requirements.txt deployment.yaml service.yaml ci-cd.yml .gitignore
SMART EDUCATION ANAL...
  .github\workflows
  ! ci-cd.yml
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  > screenshots
  .gitkeep
  > src
  .gitkeep
  > main.py
  .gitignore
  Dockerfile
  README.md
  requirements.txt

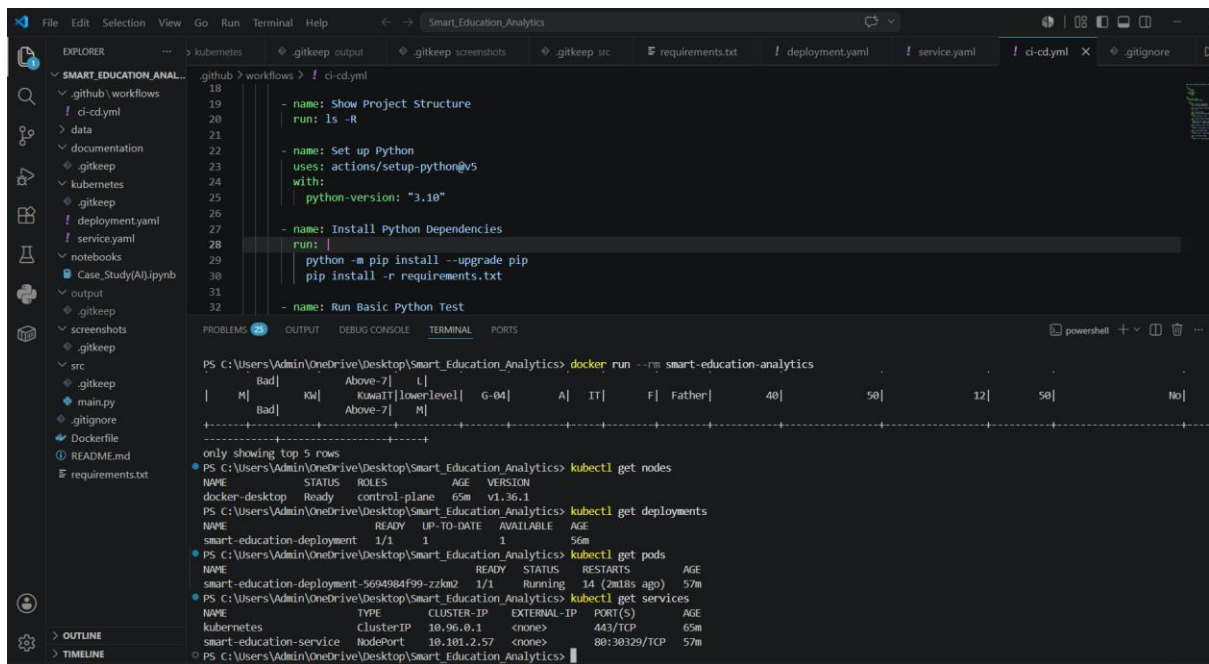
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> docker run --rm smart-education-analytics

[gender|Nationality|PlaceofBirth|StageID|GradeID|SectionID|Topic|Semester|Relation|raisedhands|VisitedResources|AnnouncementsView|Discussion|ParentAnsweringSurvey|Parentscho
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| M | Good | Kuwait | lowerlevel | G-04 | A | IT | F | Father | 15 | 16 | 2 | 20 | Yes |
| M | Good | Kuwait | lowerlevel | G-04 | A | IT | F | Father | 20 | 20 | 3 | 25 | Yes |
| M | Good | Kuwait | lowerlevel | G-04 | A | IT | F | Father | 10 | 7 | 0 | 30 | No |
| M | Bad | Kuwait | Above-7 | G-04 | A | IT | F | Father | 30 | 25 | 5 | 35 | No |
| M | Bad | Kuwait | Above-7 | G-04 | A | IT | F | Father | 40 | 50 | 12 | 50 | No |

only showing top 5 rows
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics>
```

3.Kubernetes Outputs:-



The screenshot shows a VS Code editor with a GitHub Actions workflow file named `ci-cd.yml` in the `workflows` directory. The workflow includes steps for showing project structure, setting up Python, installing dependencies, and running a basic Python test. Below the editor, the terminal displays the output of running `docker run --rm smart-education-analytics`, which shows a table of data. Subsequent terminal commands use `kubectl` to get nodes, deployments, pods, and services.

```
name: Show Project Structure
run: ls -R

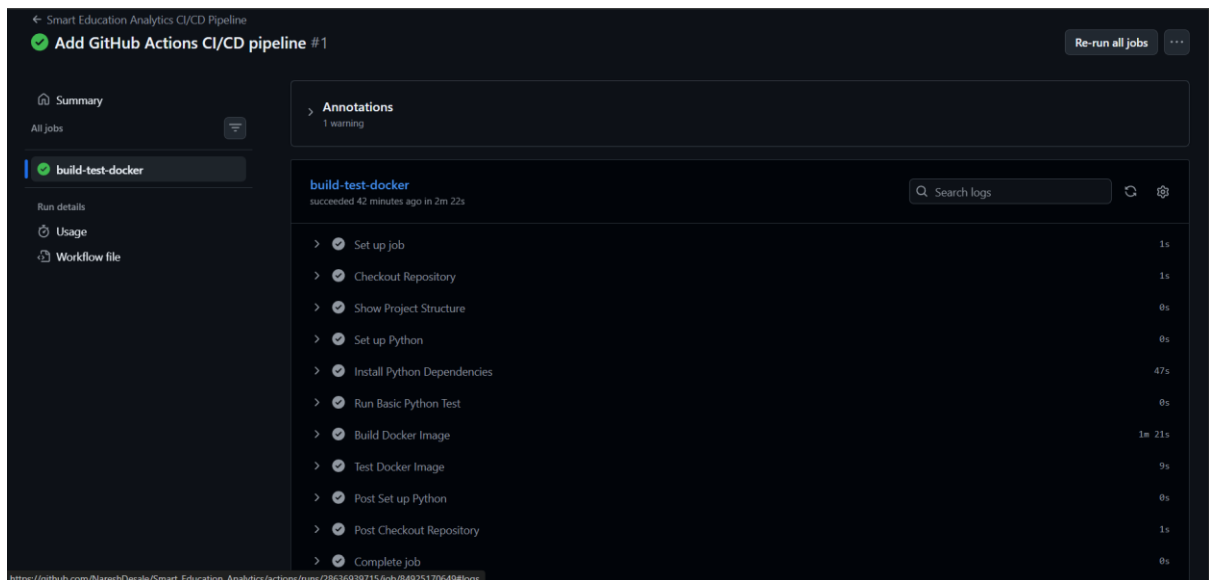
name: Set up Python
uses: actions/setup-python@v5
with:
  python-version: "3.10"

name: Install Python Dependencies
run: |
  python -m pip install --upgrade pip
  pip install -r requirements.txt

name: Run Basic Python Test
```

```
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> docker run --rm smart-education-analytics
| M|      Kw|      Above-7| L|      Above-7| LowerLevel| G-04|      A| IT|      F| Father|      40|      50|      12|      50|      No|
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+
only showing top 5 rows
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> kubectl get nodes
NAME                STATUS  ROLES    AGE   VERSION
docker-desktop      Ready   control-plane   65m   v1.36.1
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> kubectl get deployments
NAME                READY  UP-TO-DATE  AVAILABLE  AGE
smart-education-deployment  1/1    1            1          56m
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> kubectl get pods
NAME                READY  STATUS   RESTARTS  AGE
smart-education-deployment-5694984f99-zzkx2  1/1    Running   1a (2m18s ago)  57m
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics> kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
kubernetes          ClusterIP   10.96.0.1      <none>       443/TCP    65m
smart-education-service  NodePort    10.101.2.57    <none>       80:30329/TCP  57m
PS C:\Users\Admin\OneDrive\Desktop\Smart_Education_Analytics>
```

4.DevOps Pipeline:-



The screenshot shows the GitHub Actions CI/CD Pipeline interface for the `Smart Education Analytics C/CD Pipeline`. The pipeline is named `Add GitHub Actions C/CD pipeline #1` and is currently in a `build-test-docker` job. The job is successful and took 42 minutes to complete. The interface shows a list of jobs, with `build-test-docker` selected. The job details show a list of steps, including `Set up job`, `Checkout Repository`, `Show Project Structure`, `Set up Python`, `Install Python Dependencies`, `Run Basic Python Test`, `Build Docker Image`, `Test Docker Image`, `Post Set up Python`, `Post Checkout Repository`, and `Complete job`.

Summary

All jobs

Run details

Usage

Workflow file

Annotations

1 warning

build-test-docker

succeeded 42 minutes ago in 2m 22s

Search logs

Set up job 1s

Checkout Repository 1s

Show Project Structure 0s

Set up Python 0s

Install Python Dependencies 47s

Run Basic Python Test 0s

Build Docker Image 1m 21s

Test Docker Image 9s

Post Set up Python 0s

Post Checkout Repository 1s

Complete job 0s

<https://github.com/NareshDesale/Smart-Education-Analytics/actions/runs/28536939715/job/84925170649#jobs>